DAY 25 | PRE-READ

Tools, Flows and Loops

A light read before Day 25. About 15 minutes.

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Why read this?

Tomorrow AI moves from talking to doing. You will let a model use real functions, such as looking up an order, and design multi-step workflows. You will also practise systems thinking, a way of seeing why problems keep coming back.

1. Giving AI a toolbox

A receptionist cannot answer "Where is my parcel?" from memory, but can look it up in the tracking system. Tool calling gives an AI model the same ability. You describe the tools available, the model asks to use one, and **your program** runs it and hands back the result. The model never presses the buttons itself.

> **Key idea:** the model suggests, your code decides. Risky actions such as refunds should always need a human yes.

2. Workflows with decisions and loops

Real processes are not straight lines. A loan application might go back for more documents, or be escalated to a manager. **LangGraph** lets you draw an AI workflow as boxes and arrows, including "if this, go there" and "try again" loops.

3. Seeing the whole system

Traffic jams are a classic example. Building a new road can attract more cars, and the jams return. This is a **feedback loop**. Systems thinking looks for these loops, for delays between cause and effect, and for side effects, before choosing a fix.

Loop type	Everyday example
Reinforcing (snowball)	Popular restaurants get more reviews, which makes them more popular
Balancing (thermostat)	A room heats up, the heater switches off, it cools, the heater switches on

4. One-on-one feedback

Tomorrow you also meet your mentor for personal feedback. Note one thing you are proud of and one thing you want help with.

5. Think about these before class

- Which actions in your work should never be done by AI without a human check?
- Draw your morning routine as boxes and arrows. Where are the decisions?
- Think of a fix at work that created a new problem. What loop was involved?

Nagare (流れ): flow